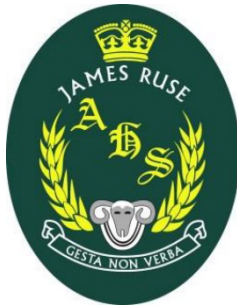


Student Name: _____

Maths class: _____



James Ruse Agricultural High School

2021 YEAR 12 Task 3 Examination

Mathematics Extension 1

General Instructions

Total marks: 40

- Reading time – 5 minutes
- Working time – 1 hour
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- In Questions 6–12, show relevant mathematical reasoning and/ or calculations

Section I – 5 marks (pages 2–3)

- Attempt Questions 1–5
- Allow about 7 minutes for this section

Section II – 35 marks (pages 4–7)

- Attempt Questions 6–12
- Allow about 53 minutes for this section

Section I

5 marks

Attempt Questions 1–5

Allow about 7 minutes for this section

Use the multiple-choice answer sheet for Questions 1–5.

1. P, Q and R are three collinear points with position vectors $\underline{p}$, $\underline{q}$ and $\underline{r}$ respectively, where Q lies between P and Q. If $|\overrightarrow{QR}| = \frac{1}{2} |\overrightarrow{PQ}|$, then $\underline{r}$ is equal to:

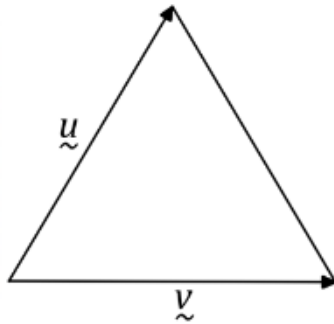
(A) $\frac{3}{2}\underline{q} - \frac{1}{2}\underline{p}$

(B) $\frac{3}{2}\underline{p} + \frac{1}{2}\underline{q}$

(C) $\frac{3}{2}\underline{q} - \frac{3}{2}\underline{p}$

(D) $\frac{1}{2}\underline{p} - \frac{3}{2}\underline{q}$

2. An equilateral triangle of side length 4 units is shown below.



Which of the following is the value of $\underline{u} \cdot \underline{v}$?

(A) 4

(B) $4\sqrt{3}$

(C) $8\sqrt{3}$

(D) 8

3. Which of the following is a well-defined expression which can be calculated involving dot-products for 2-dimensional vectors $\underline{a}$, $\underline{b}$ and $\underline{c}$?

- (A) $(\underline{a} \cdot \underline{b}) \cdot (\underline{b} \cdot \underline{a})$
- (B) $(\underline{a} \cdot \underline{b})\underline{c}$
- (C) $\underline{a} \cdot (\underline{b}\underline{c})$
- (D) $(\underline{a} \cdot \underline{b}) \cdot \underline{c}$

4. The coefficient of $\frac{1}{x}$ in the expansion of $(1+x)^n(1+\frac{1}{x})^n$ is

- (A) $\frac{n!}{(n-1)!(n+1)!}$
- (B) $\frac{(2n)!}{(n-1)!(n+1)!}$
- (C) $\frac{(2n)!}{(2n-1)!(2n+1)!}$
- (D) $\frac{(n+1)!}{2n!(n-1)!}$

5. Let $\underline{p}$, $\underline{q}$, $\underline{r}$ and $\underline{s}$ be four distinct vectors with $\underline{r} \parallel \underline{p}$ and $\underline{s} \parallel \underline{q}$. Which of the following is not an equivalent expression for the vector projection of $\underline{p}$ onto $\underline{q}$?

- | | |
|--|--|
| (A) $\left(\frac{\underline{p} \cdot \underline{q}}{ \underline{q} ^2}\right) \underline{q}$ | (C) $\left(\frac{\underline{p} \cdot \underline{s}}{ \underline{s} ^2}\right) \underline{s}$ |
| (B) $\left(\frac{\underline{p} \cdot \underline{q}}{ \underline{q} }\right) \hat{\underline{q}}$ | (D) $\left(\frac{\underline{r} \cdot \underline{s}}{ \underline{s} ^2}\right) \underline{s}$ |

Section II begins on the next page

Section II

50 marks

Attempt Questions 6-12

Allow about 53 minutes for this section

Start each question on a new page. Extra paper is available.

In Questions 6-10, your responses should include relevant mathematical reasoning and/ or calculations.

Question 6 (5 marks) Start a new page.

- (a) The point A has position vector $5\mathbf{i} - 4\mathbf{j}$ and the point B has position vector $-5\mathbf{i} - 2\mathbf{j}$.
- (i) Find the vector $\overrightarrow{AB}$ 1
 - (ii) Find $|\overrightarrow{AB}|$. Give your answer as a simplified surd. 1
- (b) Let P be the point $(5, -8)$ in the cartesian plane.
- (i) Draw the position vector $\overrightarrow{OP}$ 1
 - (ii) Write $\overrightarrow{OP}$ as a column vector. 1
 - (iii) Find a unit vector in the direction of $\overrightarrow{OP}$. 1

Question 7 (5 marks) Start a new page.

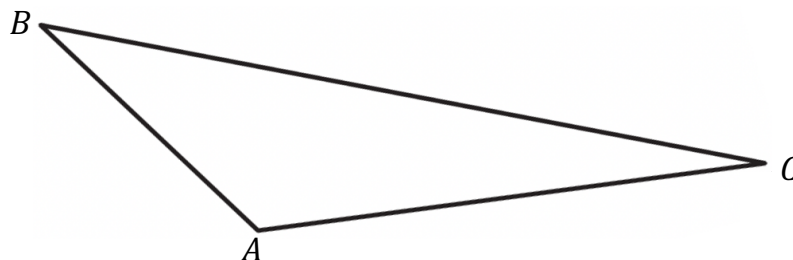
- (a) Find the angle between the vectors $3\mathbf{i} + \mathbf{j}$ and $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$. 2
- (b) Show that the vectors $\begin{pmatrix} 1 \\ 4 \end{pmatrix}$ and $\begin{pmatrix} 8 \\ -1 \end{pmatrix}$ are perpendicular. 1
- (c) Find a vector of magnitude 15 in the direction of the vector $5\mathbf{i} + 12\mathbf{j}$. 2

Please turn over

Question 8 (5 marks) Start a new page.

- (a) In the triangle ABC below, $\overrightarrow{AB} = -2\mathbf{i} + 7\mathbf{j}$ and $\overrightarrow{AC} = 12\mathbf{i} - \mathbf{j}$.

2



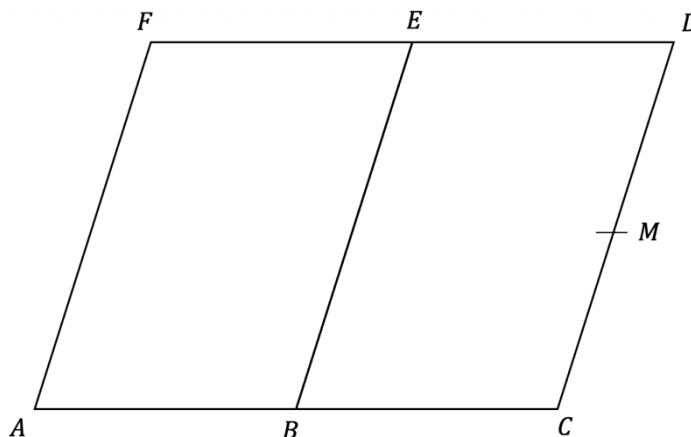
If angle BAC is 101.1° , find the area of triangle ABC ?

- (b) The coefficients of the $(r - 5)^{th}$ and $(2r - 1)^{th}$ terms in the expansion of $(1 + x)^{34}$ are equal. Find the value of r .

3

Question 9 (5 marks) Start a new page.

- (a) The diagram below shows two identical parallelograms, $AFEB$ and $BEDC$. M is the midpoint of CD .



Let the vector $\overrightarrow{AF} = \mathbf{q}$ and $\overrightarrow{AB} = \mathbf{p}$.

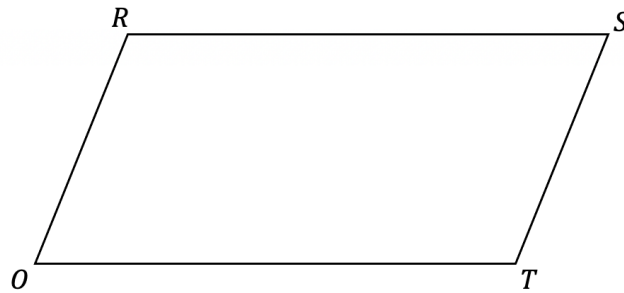
- (i) Use vector methods to show that BM is parallel to AD .
 (ii) Write $\overrightarrow{BM}$ in terms of $\mathbf{p}$ and $\mathbf{q}$.

2

1

Question 9 continues on the next page

- (b) In parallelogram $ORST$, it is known that $\overrightarrow{OR} = \underline{a} + 2\underline{b}$ and that $\overrightarrow{OT} = \underline{a} - 2\underline{b}$, where $\underline{a}$ is the position vector of the point A and $\underline{b}$ is the position vector of the point B . 2



Copy the diagram onto your answer sheet and show the positions of the points A and B .

Question 10 (5 marks) **Start a new page.**

Let X be a continuous random variable and $\lambda > 0$ be a positive real number. X is said to have a Sine Square distribution, if the probability density function of X is expressed as:

$$f_X(x) = \begin{cases} \frac{2}{\lambda\pi} \sin^2 \frac{x}{2\lambda} & \text{for } 0 < x < \lambda\pi, \text{ and } \lambda > 0; \\ 0 & \text{otherwise} \end{cases}$$

Note that $f_X(x) \geq 0$ for all values of X in the domain $(0, \lambda\pi)$.

- (a) Find the cumulative distribution function $F_X(x)$ of the Sine Square distribution. 4
- (b) Hence, or otherwise, find $P(X \leq \pi)$ when $\lambda = 2$. Give your answer to 3 decimal places. 1

Please turn over

Question 11 (5 marks) Start a new page.

(a) Find λ when the magnitude of the vector projection of $\underline{a} = \lambda \underline{i} + \underline{j}$ onto $\underline{b} = 8\underline{i} + 6\underline{j}$ equals 4. 2

(b) By considering the expansion of $(1 + x)^n$, show that 3

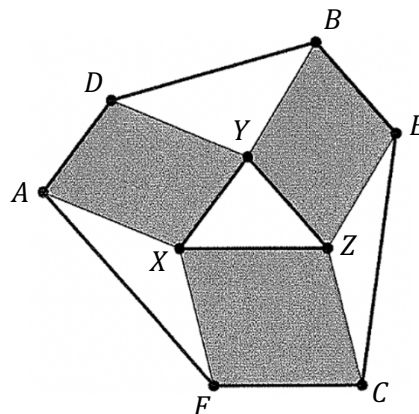
$$2 \binom{n}{2} + 6 \binom{n}{3} + 12 \binom{n}{4} + \dots + n(n-1) \binom{n}{n} = n(n-1)2^{n-2}$$

Question 12 (5 marks) Start a new page.

(a) Suppose that A, B, C, D, E, F are any points in the plane. Use vector arithmetic to verify that 2

$$\overrightarrow{AD} + \overrightarrow{BE} + \overrightarrow{CF} = \overrightarrow{BD} + \overrightarrow{CE} + \overrightarrow{AF}$$

(b) Suppose now that A, B, C, D, E, F are arranged to form the vertices of a hexagon surrounding a triangle XYZ , as shown in the diagram below. It is given that the shaded regions are parallelograms. 3



Show that the line segments DB , EC and FA can be moved parallel to themselves to assemble a triangle.

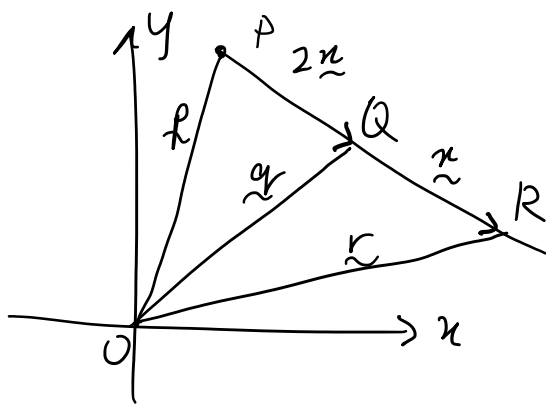
END OF EXAMINATION

Multiple choice

Sunday, 6 June 2021

9:58 pm

1.



$$\begin{aligned} \vec{r} &= \vec{OR} = \vec{OP} + \vec{PQ} + \vec{QR} \\ &= \vec{p} + \vec{PO} + \vec{OQ} + \vec{QR} \\ &= \vec{p} + -\vec{p} + \vec{q} + \frac{1}{2}(\vec{p} + \vec{q}) \\ &= -\frac{1}{2}\vec{p} + \frac{3}{2}\vec{q} \end{aligned}$$

(A)

2. $\vec{u} \cdot \vec{v} = |\vec{u}| |\vec{v}| \cos \theta$

$$= 4 \times 4 \times \cos 60^\circ$$

$$= 16 \times \frac{1}{2}$$

$$= 8$$

(D)

3.

~~A~~
(B)
~~C~~
~~D~~

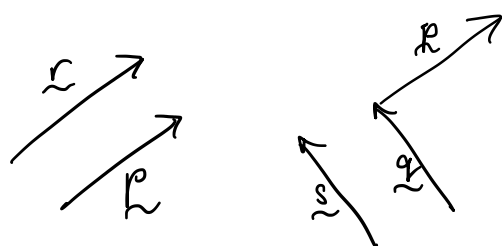
4.

$$\begin{aligned} (1+x)^n (1+\frac{1}{x})^n &= \frac{(1+x)^n (1+x)^n}{x^n} \\ &= \frac{(1+x)^{2n}}{x^n} \end{aligned}$$

$\therefore$ coeff of $\frac{1}{x}$ in $(1+x)^n (1+\frac{1}{x})^n =$ coeff of x^{n-1} in $(1+x)^{2n}$

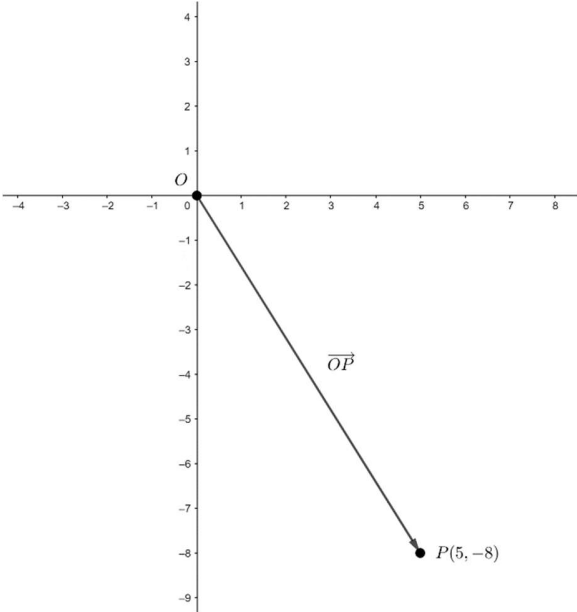
$$= \binom{2n}{n-1} = \frac{(2n)!}{(n-1)!(n+1)!} \quad \text{(B)}$$

5.



$$\frac{\vec{p} \cdot \vec{q}}{|\vec{q}|^2} \vec{q}$$

(D)

Year 12 Extension 1: Q6		
Suggested Solutions	Marks Awarded	Marker's Comments
(a) Subtracting $\overrightarrow{OB}$ from $\overrightarrow{OA}$: (i) $\overrightarrow{AB} = (-5 - 5)\underline{i} + (-2 - (-4))\underline{j} = -10\underline{i} + 2\underline{j}$ (ii) $ \begin{aligned} \overrightarrow{AB} &= \sqrt{(-10)^2 + 2^2} \\ &= \sqrt{104} \\ &= 2\sqrt{26} \end{aligned} $	1	Handled well.
(b) (i)  (ii) $\overrightarrow{OP} = \begin{bmatrix} 5 \\ -8 \end{bmatrix}$	1	Some students failed to give reference/place the vector in the Cartesian plane. Handled well, except for notation. Vectors are not points, so rounded brackets shouldn't be used. For example,
(iii) Multiply by the reciprocal of the magnitude of $\overrightarrow{OP}$ to obtain a vector in the direction of $\overrightarrow{OP}$ with magnitude 1: $ \overrightarrow{OP} = \sqrt{5^2 + (-8)^2} = \sqrt{25 + 64} = \sqrt{89}$ Hence the required unit vector is $\frac{1}{\sqrt{89}} \begin{bmatrix} 5 \\ -8 \end{bmatrix}$	1	(5, -8) is notation for a point. Why is this important? Because in the context of vectors, points AND vectors are often mixing together as part of a problem. The two must be separated in notation to avoid confusion.

Question 7 (5 marks)

- a) Find the angle between the vectors $3\mathbf{i}+\mathbf{j}$ and $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$. (2 marks)
- b) Show that the vector $\begin{pmatrix} 1 \\ 4 \end{pmatrix}$ and $\begin{pmatrix} 8 \\ -2 \end{pmatrix}$ are perpendicular. (1 mark)
- c) Find a vector of magnitude 15 in the direction of the vector $5\mathbf{i}+12\mathbf{j}$. (2 marks)

Part (a) Solution

Find the angle between the vectors $3\mathbf{i}+\mathbf{j}$ and $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$.

$$\begin{pmatrix} 3 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \left| \begin{pmatrix} 3 \\ 1 \end{pmatrix} \right| \left| \begin{pmatrix} 2 \\ -1 \end{pmatrix} \right| \cos \theta$$

(1 mark)

$$5 = \sqrt{10} \times \sqrt{5} \cos \theta$$

$$\cos \theta = \frac{\sqrt{5}}{\sqrt{10}} = \frac{1}{\sqrt{2}}$$

$$\theta = 45^\circ$$

(1 mark)

Part (b) Solution

Show that the vector $\begin{pmatrix} 1 \\ 4 \end{pmatrix}$ and $\begin{pmatrix} 8 \\ -2 \end{pmatrix}$ are perpendicular.

$$\begin{aligned} \begin{pmatrix} 1 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} 8 \\ -2 \end{pmatrix} &= 8 - 8 \\ &= 0 \end{aligned}$$

Since the dot product is 0, the vectors are perpendicular.

(1 mark)

Part (c) Solution

Find a vector of magnitude 15 in the direction of the vector $5\mathbf{j} + 12\mathbf{j}$.

$$|5\mathbf{j} + 12\mathbf{j}| = 13 \quad (1 \text{ mark})$$

$$\begin{array}{l} \text{The vector of magnitude 15} \\ \text{and in the direction of vector} \\ 5\hat{\mathbf{i}} + 12\hat{\mathbf{j}} \end{array} = \frac{15}{13}(5\mathbf{j} + 12\mathbf{j}) \quad (1 \text{ mark})$$

MATHEMATICS Extension 1 : Question... 8....

Suggested Solutions

Marks

Marker's Comments

a)

$$|\vec{AB}| = \sqrt{(-2)^2 + (7)^2} = \sqrt{53}$$

$$|\vec{AC}| = \sqrt{12^2 + (-1)^2} = \sqrt{145}$$

$$\therefore \text{Area} = \frac{\sqrt{53} \times \sqrt{145} \times \sin 101.1^\circ}{2}$$

$$= 43.01 \text{ u}^2$$

①

①

$$b) (1+x)^{34} = {}^{34}C_0 + {}^{34}C_1 x + \dots + {}^{34}C_r x^r + \dots + {}^{34}C_{34} x^{34}$$

$\uparrow$ 1st term $\uparrow$ 2nd term

$$\therefore \text{Coefficient of } (r-5)^{\text{th}} \text{ term} = {}^{34}C_{r-6}$$

$$\text{coefficient of } (2r-1)^{\text{th}} \text{ term} = {}^{34}C_{2r-2}$$

$$\therefore r-6 = 2r-2 \text{ or } (r-6) + (2r-2) = 34$$

$$r = -4$$

$$3r - 8 = 34$$

$$\text{but } r \geq 0.$$

$$3r = 42$$

$$r = 14$$

①

①

①

$$\therefore r = 14 \text{ only.}$$

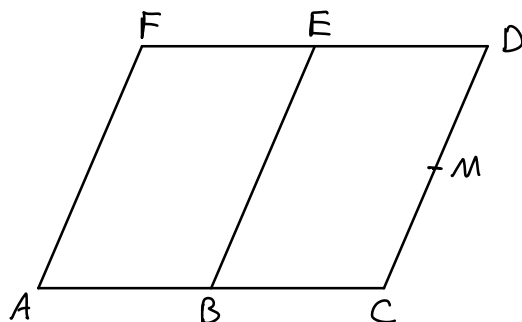
Note: If your answer is $\frac{40}{3}$, you will receive 1/3 MAX!

- Getting $r=14$ is only worth 1 mark on its own without appropriate working

- You will also not receive full marks if your final step was trial and error but you did not sub the value to check that it is correct.

Question 9:

ai,



$$\vec{AF} = z$$

$$\vec{AB} = r$$

RTP: $AD \parallel BM$

$$\vec{AD} = \vec{AC} + \vec{CD}$$

$$= 2\vec{AB} + \vec{CD}$$

$$\vec{AD} = 2r + z$$

①

$$\vec{BM} = \vec{BC} + \vec{CM}$$

$$\vec{BM} = r + \frac{1}{2}z$$

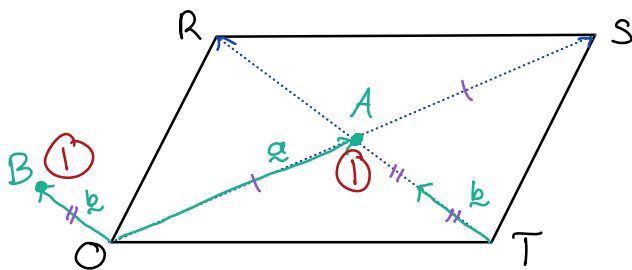
$$= \frac{1}{2}(2r + z)$$

$$\textcircled{1} \left\{ \begin{array}{l} \vec{BM} = \frac{1}{2}\vec{AD} \\ \therefore \vec{BM} \text{ is a scalar multiple of } \vec{AD}, \text{ then } BM \parallel AD. \end{array} \right.$$

ii, From above, $\vec{BM} = r + \frac{1}{2}z$

①

b,



$$\vec{OR} = a + 2b \quad \textcircled{1}$$

$$\vec{OT} = a - 2b \quad \textcircled{2}$$

$$\textcircled{1} + \textcircled{2}: 2a = \vec{OR} + \vec{OT}$$

$$a = \frac{1}{2}(\vec{OR} + \vec{OT})$$

$$\textcircled{1} - \textcircled{2}: 4b = \vec{OR} - \vec{OT}$$

$$b = \frac{1}{4}(\vec{OR} - \vec{OT})$$

MATHEMATICS Extension 1 : Question...10...

Suggested Solutions	Marks	Marker's Comments
a) $\int_0^x \frac{2}{\lambda\pi} \sin^2 \frac{x}{2\lambda} dx$ $= \frac{2}{\lambda\pi} \int_0^x \frac{1}{2} (1 - \cos \frac{x}{\lambda}) dx$ $= \frac{1}{\lambda\pi} \int_0^x (1 - \cos \frac{x}{\lambda}) dx$ $= \frac{1}{\lambda\pi} \left[x - \lambda \sin \frac{x}{\lambda} \right]_0^x$ $= \frac{1}{\lambda\pi} \left[x - \lambda \sin \frac{x}{\lambda} \right] - 0$ $\therefore$ $F_x(x) = \begin{cases} 0, & x \leq 0 \\ \frac{1}{\lambda\pi} \left(x - \lambda \sin \frac{x}{\lambda} \right), & 0 < x < \lambda\pi \\ 1, & x \geq \lambda\pi \end{cases}$	1 1 1 1	setting up the integral with boundaries. If not using boundaries, first mark will go towards working out the + c. must show all three parts. Not many students did this.
b) when $\lambda=2$, $x=\pi$ $P(x \leq \pi) = \frac{1}{2\pi} \left(\pi - 2 \sin \frac{\pi}{2} \right)$ $= \frac{\pi}{2\pi} - \frac{2}{2\pi}$ $= \frac{1}{2} - \frac{1}{\pi}$ $= 0.182 \text{ (3dp)}$	1	$\frac{1}{2} - \frac{1}{\pi}$ was also accepted. The sin value must be evaluated. A lot of students used degrees rather than radians!

Question 11

Sunday, 8 August 2021 10:15 pm

(a) Find λ when the magnitude of the vector projection of $\underline{a} = \lambda\hat{i} + \hat{j}$ onto $\underline{b} = 8\hat{i} + 6\hat{j}$ equals 4.

2

Solution:

$$\begin{aligned} |\text{proj}_{\underline{b}} \underline{a}| &= \left| \frac{\underline{a} \cdot \underline{b}}{|\underline{b}|} \right| \\ &= \left| \frac{\begin{pmatrix} \lambda \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 8 \\ 6 \end{pmatrix}}{\sqrt{8^2 + 6^2}} \right| \\ &= \left| \frac{8\lambda + 6}{10} \right| = 4 \end{aligned}$$

$$8\lambda + 6 = 40 \quad \text{or} \quad 8\lambda + 6 = -40$$

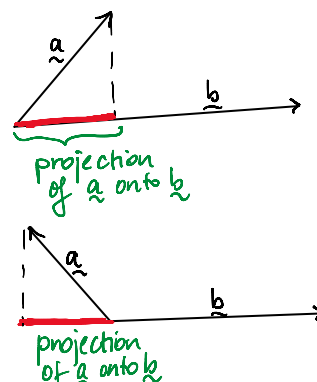
$$8\lambda = 34 \quad \quad \quad 8\lambda = -46$$

$$\lambda = 4.25 \quad \text{or} \quad -5.75$$

①

①

Many student did not consider the magnitude of the vector projection. They omitted the absolute value and so only found one value of λ .



1 mark was awarded for only one value of λ .

2 marks were awarded for two correct values of λ .

(b) By considering the expansion of $(1+x)^n$, show that

3

$$2 \binom{n}{2} + 6 \binom{n}{3} + 12 \binom{n}{4} + \dots + n(n-1) \binom{n}{n} = n(n-1)2^{n-2}$$

Solution:

$$(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \binom{n}{3}x^3 + \binom{n}{4}x^4 + \dots + \binom{n}{n}x^n$$

Differentiating both sides:

$$n(1+x)^{n-1} = \binom{n}{1} + 2\binom{n}{2}x + 3\binom{n}{3}x^2 + 4\binom{n}{4}x^3 + \dots + n\binom{n}{n}x^{n-1}$$

Differentiating both sides again:

$$n(n-1)(1+x)^{n-2} = 2\binom{n}{2} + 6\binom{n}{3}x + 12\binom{n}{4}x^2 + \dots + n(n-1)\binom{n}{n}x^{n-2}$$

Substituting $x=1$ in both sides

$$n(n-1)2^{n-2} = 2\binom{n}{2} + 6\binom{n}{3} + 12\binom{n}{4} + \dots + n(n-1)\binom{n}{n}$$

1st mark was awarded for differentiating both sides once with explanation

2nd mark was awarded for differentiating both sides again with explanation

3rd mark was awarded for substitution of $x=1$ into the expression with explanation

Notes:

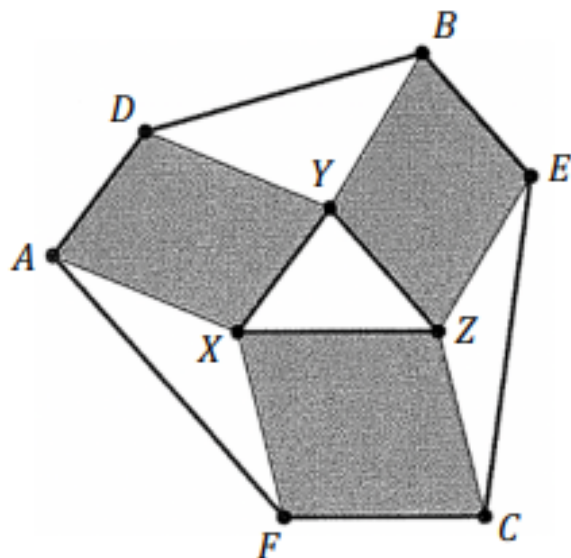
- Many students did not write out enough terms in their initial statement so were unable to show the final statement given in the question.
- Students wrote the extra terms in subsequent lines of working without showing where they initially came from.
- Students who did not show enough terms to arrive at the given expression were awarded maximum 2 marks
- Many students wrote lines of working without explaining how they arrived at each line.
- Students who did not explain they were differentiating each line were awarded only the mark for substitution.
- Many students who used unacceptable notation and explanations were NOT awarded marks. These included: LHS', RHS', $\frac{d}{dx}$, $((1+x)^n)'$, "diff"

Question 12 (5 marks) **Start a new page.**

- (a) Suppose that A, B, C, D, E, F are any points in the plane. Use vector arithmetic to verify that **2**

$$\overrightarrow{AD} + \overrightarrow{BE} + \overrightarrow{CF} = \overrightarrow{BD} + \overrightarrow{CE} + \overrightarrow{AF}$$

- (b) Suppose now that A, B, C, D, E, F are arranged to form the vertices of a hexagon surrounding a triangle XYZ , as shown in the diagram below. It is given that the shaded regions are parallelograms. **3**



Show that the line segments DB , EC and FA can be moved parallel to themselves to assemble a triangle.

(a) Suppose that A, B, C, D, E, F are any points in the plane. Use vector arithmetic to verify that

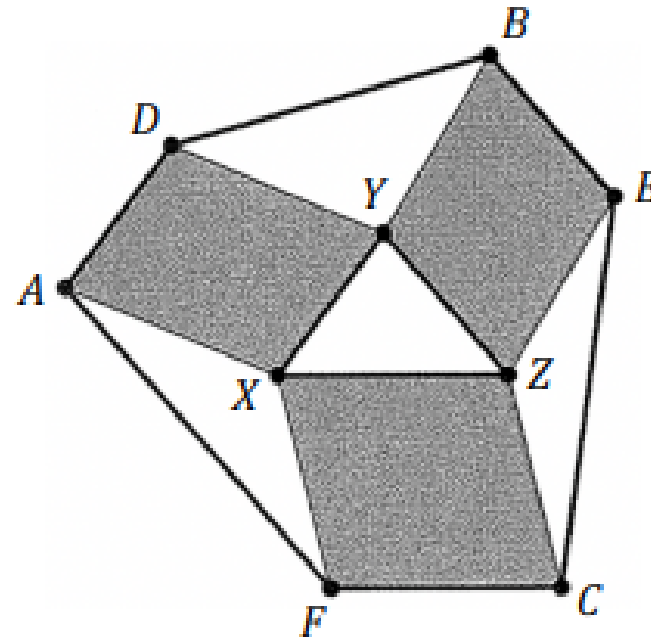
2

$$\overrightarrow{AD} + \overrightarrow{BE} + \overrightarrow{CF} = \overrightarrow{BD} + \overrightarrow{CE} + \overrightarrow{AF}$$

$$\begin{aligned} & \overrightarrow{AD} + \overrightarrow{BE} + \overrightarrow{CF} \\ = & \overrightarrow{AF} + \overrightarrow{FD} + \overrightarrow{BD} + \overrightarrow{DE} + \overrightarrow{CE} + \overrightarrow{EF} & 1 \\ = & \overrightarrow{AF} + \overrightarrow{BD} + \overrightarrow{CE} + \overrightarrow{EF} + \overrightarrow{FD} + \overrightarrow{DE} \\ = & \overrightarrow{AF} + \overrightarrow{BD} + \overrightarrow{CE} + \overrightarrow{EE} & 1 \\ = & \overrightarrow{AF} + \overrightarrow{BD} + \overrightarrow{CE} \end{aligned}$$

- (b) Suppose now that A, B, C, D, E, F are arranged to form the vertices of a hexagon surrounding a triangle XYZ , as shown in the diagram below. It is given that the shaded regions are parallelograms.

3



Show that the line segments DB , EC and FA can be moved parallel to themselves to assemble a triangle.

$$\begin{aligned} & \overrightarrow{BD} + \overrightarrow{AF} + \overrightarrow{CE} \\ = & \overrightarrow{BY} + \overrightarrow{YD} + \overrightarrow{AX} + \overrightarrow{XF} + \overrightarrow{CZ} + \overrightarrow{ZE} \end{aligned} \quad 1$$

$\overrightarrow{BY} = \overrightarrow{EZ}$ Since BYZE is a parallelogram, and opposite sides are parallel and equal.

$\overrightarrow{YD} = \overrightarrow{XA}$ Since ADYX is a parallelogram, and opposite sides are parallel and equal 1

$\overrightarrow{XF} = \overrightarrow{ZC}$ Since XFCZ is a parallelogram, and opposite sides are parallel and equal

Now using The above informations, we obtain:

$$\begin{aligned} & \overrightarrow{BY} + \overrightarrow{YD} + \overrightarrow{AX} + \overrightarrow{XF} + \overrightarrow{CZ} + \overrightarrow{ZE} \\ = & \overrightarrow{BY} + \overrightarrow{YD} + \overrightarrow{XF} - \overrightarrow{YD} - \overrightarrow{XF} - \overrightarrow{BY} \\ = & \overrightarrow{BY} - \overrightarrow{BY} + \overrightarrow{YD} - \overrightarrow{YD} + \overrightarrow{XF} - \overrightarrow{XF} \\ = & \vec{0} \end{aligned} \quad 1$$

Therefore the three vectors $\overrightarrow{DB}$, $\overrightarrow{EC}$ and $\overrightarrow{FA}$ are the sides of a triangle.